

Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty

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Chapter overview

Chapters 7 and 8 fitted lines and additive models and read them as conditional comparisons. Along the way they used tools they could only name: a factor coded as 0/1 columns, a contrast between two categories, a joint test of several added coefficients, an information criterion, a squared predictor, a logged outcome, a covariance estimate that allows unequal spread, and a covariance that respects pupils sitting in schools. This chapter develops each of them once, on the same 98 occupation records, and then on Project STAR, where a randomized design supplies a different warrant. By the end of the chapter you will be able to code and read a factor with several levels, estimate a contrast with the right uncertainty, compare nested models and read AIC and BIC, change a predictor's form or an outcome's scale and say what question changed, choose between classical, HC3, cluster-robust, and bootstrap uncertainty for a stated reason, and say what none of these calculations repairs.

A regression report rarely stops at one additive equation. It compares groups that are categories, asks whether a set of added variables matters jointly, ranks candidate models, lets a relationship curve or a scale change, and reports uncertainty calculated in more than one way. Each of these steps has a small mathematics of its own and a temptation of its own: reading a coefficient as “the effect of type”, adding two standard errors, comparing criteria across different outcomes, or treating a robust standard error as a repair for a design problem.

We keep the occupation records of Chapter 8, with their 98 complete rows, income in thousand 1971 Canadian dollars, and the additive model of education, prestige, percentage of women, and type, so that every tool is seen on numbers already read. Nothing here is a new study. The chapter's contribution is to give a scattered set of operations one home, one vocabulary, and one set of records, so that the difference between changing a model, changing a question, and changing an uncertainty calculation stays visible. The chapter closes with Project STAR, where the uncertainty calculation has to follow the design that produced the data.

Each visible R example follows from the earlier visible preparation in this chapter; Chapters 7 and 8 supply familiarity, not an existing R session. The type levels remain *bc*, *prof*, and *wc*, the full additive formula remains `income_k ~ education + prestige + women + type`, and every adjusted type comparison names education, prestige, and the percentage of women as its conditioning set.

1 Predictors That Are Categories

1.1 One indicator, then several

Chapter 7, Section 3.3, fitted a single 0/1 predictor to 24 invented feedback-by-practice scores. The fitted line passed through the two group means, its slope was the difference between them, 4.0000 with SE 1.6967 and $t = 2.3575$ on 22 df, and that test was the equal-variance two-sample test written as a regression. Nothing about that arrangement is special to two groups.

A factor with J levels enters a regression as $J - 1$ indicator columns. One level is omitted and becomes the **reference**: the intercept is its fitted mean, and each indicator coefficient is a difference from it. With three occupation types and blue-collar as the reference,

$$\hat{y}_g = \hat{\alpha} + \hat{\delta}_{\text{prof}} D_{\text{prof}} + \hat{\delta}_{\text{wc}} D_{\text{wc}}, \quad \hat{\mu}_{\text{bc}} = \hat{\alpha}, \quad \hat{\mu}_{\text{prof}} = \hat{\alpha} + \hat{\delta}_{\text{prof}}, \quad \hat{\mu}_{\text{wc}} = \hat{\alpha} + \hat{\delta}_{\text{wc}}.$$

Two indicators, not three, carry three means: the third column would be the intercept column minus the other two, and the design matrix would lose full rank. The choice of reference is a choice of coordinates, not a choice of model.

We re-create the Chapter 8 objects visibly so that every later calculation in this chapter has a stated origin. The 98 rows are the type-complete occupations; `m0`, `m1`, and `m2` are the education-only, the education-and-prestige, and the full additive models.

```
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
d <- prestige[!is.na(prestige$type), ]
d$type <- factor(d$type, levels = c("bc", "prof", "wc"))
m0 <- lm(income_k ~ education, data = d)
m1 <- lm(income_k ~ education + prestige, data = d)
m2 <- lm(income_k ~ education + prestige + women + type, data = d)
c(rows = nrow(d), residual_df = df.residual(m2))
```

rows	residual_df
98	92

The 98 retained records hold 44 blue-collar, 31 professional, and 23 white-collar occupations. The full model fits six coefficients, leaving 92 residual degrees of freedom. Complete-case selection is how these rows were chosen; it is not random sampling from a labour market (Fox et al., 2022; Fox & Weisberg, 2019).

1.2 Indicators recover fitted group means

Before adjusting for anything, fit income on type alone. Printing the model matrix for one row of each type makes the coding visible rather than assumed, and placing the fitted values beside the observed group means shows what the indicators are doing.

```
type_levels <- data.frame(type = factor(c("bc", "prof", "wc"),
  levels = levels(d$type)))
model.matrix(~ type, data = type_levels)
```

```
      (Intercept) typeprof typewc
1             1           0       0
2             1           1       0
3             1           0       1
attr(,"assign")
[1] 0 1 1
attr(,"contrasts")
attr(,"contrasts")$type
[1] "contr.treatment"
```

```
group_fit <- lm(income_k ~ type, data = d)
cbind(type_levels,
  observed = as.numeric(tapply(d$income_k, d$type, mean)),
  fitted = predict(group_fit, type_levels))
```

```
   type observed   fitted
1  bc  5.374136  5.374136
2 prof 10.559452 10.559452
3  wc  5.052304  5.052304
```

```
coef(group_fit)
```

```
      (Intercept)      typeprof      typewc
      5.374136      5.185315     -0.321832
```

The blue-collar row carries zeros in both indicator columns, so its fitted value is the intercept alone. The observed means are 5.3741, 10.5595, and 5.0523 thousand dollars, and the fitted values reproduce them exactly. The coefficients are 5.3741, 5.1853, and -0.3218 : the intercept is the blue-collar mean, and each indicator coefficient is that type's mean minus the blue-collar mean.

This is not an accident of these data. Least squares minimizes the sum of squared deviations, and with one free parameter per group the minimizing value within each group is that group's own mean. A group-only regression is therefore a description of three means with no adjustment in it at all.

Keep that description apart from the adjusted comparison of the full model. In `m2` the professional and white-collar coefficients are 0.3242 and 0.2353: they compare fitted incomes at the *same* education, prestige, and percentage of women, not the observed incomes of differently composed groups. The observed professional advantage of 5.1853 thousand dollars and the adjusted 0.3242 are answers to different questions, and neither is “the effect of occupational type”.

1.3 Releveling and centering change coordinates, not the fit

Releveling makes a different level the reference. Refit the group-only model with white-collar as reference and compare the fitted values.

```
group_fit_wc <- lm(income_k ~ relevel(type, ref = "wc"), data = d)
coef(group_fit_wc)
```

```
              (Intercept)    relevel(type, ref = "wc")bc
              5.052304                0.321832
relevel(type, ref = "wc")prof
              5.507147
```

```
max(abs(fitted(group_fit) - fitted(group_fit_wc)))
```

```
[1] 8.171241e-14
```

The coefficients are now 5.0523, 0.3218, and 5.5071: the intercept is the white-collar mean, and the two indicator coefficients are the blue-collar and professional means measured from it. The fitted values agree with the blue-collar coding to about 10^{-13} , the size of floating-point rounding. The same three means are being reported in a different coordinate system.

The full model behaves the same way. Releveling `m2` to white-collar gives intercept 0.6559, blue-collar -0.2353 , and professional 0.0889, while the education, prestige, and women coefficients are unchanged to the last printed digit. Centering a continuous predictor is the analogous operation: subtracting a constant moves the profile the intercept refers to, and leaves every fitted value, residual, and fixed substantive contrast alone.

Two consequences are worth keeping. First, a coefficient's size is not interpretable without the reference it is measured from; “the professional coefficient” is short for “professional minus

whichever level is currently omitted”. Second, neither releveling nor centering creates information. Chapter 8 documented how thin the joint support across types is at common education and prestige values; changing coordinates does not add a single observed comparison.

1.4 The one-way analysis of variance of Chapter 4 as a regression

Chapter 4 tested equality of several group means with a one-way analysis of variance. In the group-only regression, that null is the null that both indicator coefficients are zero, so the reduced model is the intercept-only model and the comparison is a nested-model comparison:

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/2}{RSS_{\text{full}}/(98 - 3)} = \frac{(1734.0792 - 1138.1230)/2}{1138.1230/95} = 24.8725.$$

The reduced model’s residual sum of squares is the total sum of squares about the grand mean, 1734.0792; the group-only model leaves 1138.1230 on 95 degrees of freedom. Two restrictions are tested, and the denominator is the full model’s residual mean square.

```
null_fit <- lm(income_k ~ 1, data = d)
anova(null_fit, group_fit)
```

Analysis of Variance Table

```
Model 1: income_k ~ 1
Model 2: income_k ~ type
  Res.Df    RSS Df Sum of Sq    F    Pr(>F)
1     97 1734.1
2     95 1138.1  2     595.96 24.872 2.057e-09 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
type_aov <- aov(income_k ~ type, data = d)
c(regression_F = anova(null_fit, group_fit)$F[2],
  anova_F = summary(type_aov)[[1]][["F value"]][1],
  max_fitted_difference = max(abs(fitted(group_fit) - fitted(type_aov))))
```

regression_F	anova_F	max_fitted_difference
24.87246	24.87246	0.00000

`aov()` returns the same statistic from the same fitted means, the same within-group residual sum of squares, the same two restrictions, and the same denominator degrees of freedom, and its fitted values are identical. The equality is algebraic. It holds whether or not group membership was randomized, and it therefore supplies no causal warrant of its own; what randomization would add is a reason to read the compared means as assignment effects, which Section 7 takes up.

The Chapter 4 cautions travel with the identity. An omnibus F says that the three means are not all equal under the stated reference distribution; it does not say which differ, and any comparison chosen after looking at this result still needs the family calibration of Chapter 4. The next section gives the calculation that a single declared comparison requires.

2 Contrasts and Adjusted Group Comparisons

2.1 A contrast needs coefficient covariance

A **contrast** is a declared linear combination of coefficients, $C = c'\beta$, chosen before the comparison is read. Its estimate is the same combination of the fitted coefficients, and its variance is the corresponding quadratic form in the coefficient covariance matrix:

$$\widehat{C} = c'\widehat{\beta}, \quad \widehat{\text{Var}}(\widehat{C}) = c'\widehat{\mathbf{V}}c, \quad SE(\widehat{C}) = \sqrt{c'\widehat{\mathbf{V}}c}.$$

Estimated coefficients are correlated, because the predictors they belong to are correlated. The quadratic form is exactly the bookkeeping that tracks those correlations: for a difference of two coefficients it gives $\text{Var}(b_1) + \text{Var}(b_2) - 2\text{Cov}(b_1, b_2)$. Take professional minus white-collar in `m2`, with the coefficient order intercept, education, prestige, women, professional, white-collar, so that $c = (0, 0, 0, 0, 1, -1)'$.

```

c_vec <- c(0, 0, 0, 0, 1, -1)
V <- vcov(m2)
C_hat <- sum(c_vec * coef(m2))
C_var <- drop(c_vec %*% V %*% c_vec)
c(estimate = C_hat, variance = C_var, SE = sqrt(C_var))

```

```

      estimate    variance      SE
0.08886278 1.16638502 1.07999307

```

```

C_hat + c(-1, 1) * qt(.975, df.residual(m2)) * sqrt(C_var)

```

```

[1] -2.056097  2.233822

```

The estimate is $0.3242 - 0.2353 = 0.0889$ thousand dollars. Its variance assembles the two coefficient variances and their covariance,

$$2.1406 + 0.9698 - 2(0.9720) = 1.1664,$$

so the SE is 1.0800 and, with $t_{.975,92} = 1.9861$, the interval is $[-2.0561, 2.2338]$.

Two wrong calculations are worth naming because both are common. Adding the two standard errors gives $1.4631 + 0.9848 = 2.4479$, which is not the standard error of anything here; it is an upper bound that the positive covariance between the two coefficients makes badly loose. Adding the two variances and discarding the covariance gives 3.1104 and an SE of 1.7636, also wrong. Only the quadratic form is the variance of the declared comparison.

The same machinery covers any declared comparison, because c is simply the recipe written as a vector. Setting one entry to 1 and the rest to 0 returns an ordinary coefficient and its usual SE. Setting the two type entries to 1/2 each compares the average of the two named types with the blue-collar reference. Whenever the non-intercept entries sum to zero, the combination is a contrast in the Chapter 4 sense and does not depend on which level is the reference; the comparison above is such a combination, which is why Section 1.3 found the same 0.0889 as the professional coefficient of the releveled fit. What never changes is that the variance is the quadratic form, computed from the same $\widehat{V}$ that `vcov()` returns.

Declaring the comparison in advance matters as much as computing it correctly. Chapter 4's family calibration applies to comparisons chosen after seeing an omnibus result, and a contrast selected because it looked large in the fitted coefficients is exactly such a choice. The calculation in this subsection assumes one comparison named before the model was read.

Read the interval for what it is. It is a working-model benchmark for an adjusted comparison: professional against white-collar occupations at the same education, prestige, and percentage

of women. It is not a difference of observed group means, and its width does not overcome the limited cross-type support that Chapter 8 documented. An interval can be arithmetically correct and still describe a comparison that the records barely observe.

2.2 Adjusted block tests

A contrast tests one declared combination. A **block test** asks whether a set of coefficients is jointly zero, and it is the nested-model comparison of Section 1.4 with a different reduced model. Two blocks are worth testing here, and they are different questions.

```
no_type <- lm(income_k ~ education + prestige + women, data = d)
women_type <- anova(m1, m2)
type_after <- anova(no_type, m2)
```

Adding the percentage of women and the two type indicators to `m1` tests three restrictions against the full model's 92 residual degrees of freedom. Dropping type from the full model tests two restrictions against the same denominator.

Table 1

Two declared adjusted block comparisons.

Added	df	F	p
Women and type	3	11.5088	< .0001
Type after women	2	0.0319	.9686

The first block is strongly supported: $F(3, 92) = 11.5088$ against a 5% critical value of 2.7036. The second is not: $F(2, 92) = 0.0319$ with $p = .9686$ against a critical value of 3.0954.

Neither result says what it is often read as saying. Joint evidence for the first block does not establish that each of the three added columns contributes; the percentage of women could be carrying the whole block. And the second result is not proof that occupational type is irrelevant. It says that, once education, prestige, and the percentage of women are in the equation, the two type indicators add no detectable reduction in residual sum of squares *on these 98 records*, where the types barely overlap on the other predictors. Absence of detectable joint contribution is not evidence of no difference.

Two further distinctions matter. `anova(m2)` on a single model gives **sequential** sums of squares in formula order: each line tests a term given only the terms above it, so the answers depend on the order in which the formula was written. Those lines are not order-free tests of every term, and only the last one coincides with the corresponding block test. Separately, Welch-type and robust-covariance tests are different procedures with different reference distributions, not variants of this formula; Section 5 gives the robust version of exactly this type block.

2.3 Adjusted means under a common slope: the Baumann reference

Chapter 6 introduced the analysis of covariance on a separate invented twelve-pupil example with a pretest: a group comparison made at a common baseline value under a common slope. The real-record reference for that idea belongs here, because it is the same indicator-and-contrast machinery with one continuous covariate.

Baumann and colleagues studied 68 fourth-grade pupils in one rural school under three reading-instruction conditions (Baumann et al., 1992). The `carData::Baumann` distribution holds the 66 pupils with a recorded first posttest, 22 per group. The teaching analysis uses the first pretest and first posttest only, and relabels the package's `Basal`, `DRTA`, and `Strat` as `DRA`, `DRTA`, and `Think-aloud`, with `DRA` as the reference. The model assumes one common pretest slope across groups:

$$Y_{ij} = \mu_j + \beta(X_{ij} - \bar{X}) + \varepsilon_{ij}, \quad \bar{Y}_j^{\text{adj}} = \bar{Y}_j - \hat{\beta}(\bar{X}_j - \bar{X}).$$

The adjusted mean is the observed group mean moved along the common slope by the group's pretest advantage or deficit relative to the grand pretest mean.

```
reading <- carData::Baumann
reading$group <- factor(reading$group,
  levels = c("Basal", "DRTA", "Strat"),
  labels = c("DRA", "DRTA", "Think-aloud"))
ancova <- lm(post.test.1 ~ group + pretest.1, data = reading)
summary(ancova)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.5966478	1.1845062	-0.5037101	6.162502e-01
groupDRTA	3.6265538	0.7361861	4.9261371	6.553543e-06
groupThink-aloud	2.0361644	0.7449616	2.7332475	8.161831e-03
pretest.1	0.6931872	0.1014697	6.8314735	4.205253e-09

```
reading_means <- tapply(reading$post.test.1, reading$group, mean)
pretest_means <- tapply(reading$pretest.1, reading$group, mean)
adjusted <- reading_means - coef(ancova)["pretest.1"] *
  (pretest_means - mean(reading$pretest.1))
adjusted
```

DRA	DRTA	Think-aloud
6.188185	9.814739	8.224349

```
varying_slopes <- lm(post.test.1 ~ group * pretest.1, data = reading)
anova(ancova, varying_slopes)
```

Analysis of Variance Table

Model 1: post.test.1 ~ group + pretest.1

Model 2: post.test.1 ~ group * pretest.1

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	62	365.43				
2	60	347.97	2	17.461	1.5054	0.2302

The common pretest slope is 0.6932 with SE 0.1015. Against DRA, the DRTA coefficient is 3.6266 (SE 0.7362) and the Think-aloud coefficient is 2.0362 (SE 0.7450); the residual mean square is 5.8940 on 62 degrees of freedom. The group pretest means are 10.5000, 9.7273, and 9.1364 against a grand mean of 9.7879, so DRA starts above and Think-aloud below the common reference point.

Adjustment moves the raw posttest means 6.6818, 9.7727, and 7.7727 to 6.1882, 9.8147, and 8.2243. The DRA mean falls because that group's pretest advantage of 0.7121 points is removed at the common slope; the Think-aloud mean rises for the mirror-image reason. The adjusted values answer a question the raw means do not: what these groups' posttest means would be if each had the grand pretest mean, *under the common-slope model*.

Two conditions carry that reading, and both need saying. The adjusted means need a **common reference** — here the grand pretest mean — and an **averaging population** over which the comparison is made. Averaging the fitted model over the common empirical pretest distribution returns the same three values, because the model is linear in the pretest and the same distribution is used for every group; that equality is why the simple shift formula suffices.

The common-slope condition is checkable. Comparing the additive model with the model that lets each group have its own pretest slope gives $F(2, 60) = 1.5054$, $p = .2302$: on these records, the varying-slope model does not improve the fit detectably. That is a check, not a demonstration that the slopes are equal, and it is where this chapter stops. Chapter 10 develops the varying-slope model itself as an interaction, with conditional slopes and their uncertainty.

3 Compare Nested Models

3.1 The extra-sum-of-squares statistic

Sections 1.4 and 2.2 were two instances of one statistic. Let the reduced model be obtained from the full model by setting q coefficients to zero, with both models fitted to the same records and the same outcome. Then

$$F = \frac{(RSS_R - RSS_F)/q}{RSS_F/(n-p)}, \quad F_{\text{overall}} = \frac{RegSS/(p-1)}{RSS/(n-p)}, \quad RegSS = \sum_i (\hat{y}_i - \bar{y})^2.$$

The numerator is the extra sum of squares the added block explains, per added coefficient. The denominator is the full model's residual mean square, the same s^2 that appears in every coefficient standard error. Under the normal linear model with independent common-variance errors, the ratio has an $F_{q, n-p}$ reference.

Both earlier tests are instances. The one-way identity of Section 1.4 sets $q = 2$ and $n - p = 95$, with $RSS_R = 1734.0792$ from the intercept-only model and $RSS_F = 1138.1230$ from the group-only model, giving $(595.9562/2)/(1138.1230/95) = 24.8725$. The adjusted type block of Section 2.2 sets $q = 2$ and $n - p = 92$, with $RSS_R = 631.3272$ from the model without type and $RSS_F = 630.8897$ from `m2`, giving $(0.4375/2)/(630.8897/92) = 0.0319$. The women-and-type block sets $q = 3$ with $RSS_R = 867.6537$, giving $(236.7640/3)/6.8575 = 11.5088$. The same three ingredients appear every time: how much residual sum of squares the block buys, how many coefficients it spends, and what a unit of unexplained variation costs in the fuller model.

Notice what must be held fixed for the subtraction to mean anything. The two models must use the same rows and the same outcome, or the difference of residual sums of squares is not a reduction attributable to the added block; and the reduced model must be obtainable from the fuller one by restrictions, or there is no null to test. Chapter 8's complete-case selection is why every comparison in this chapter runs on the same 98 records.

The **overall** regression test is the special case in which the reduced model is the intercept alone, so every non-intercept coefficient is restricted to zero. It is the line R prints at the foot of `summary()`.

```
summary(m2)$fstatistic
```

```
value  numdf  dendf
32.1747  5.0000 92.0000
```

```
summary(m0)$fstatistic
```

```
value  numdf  dendf
47.19561  1.00000 96.00000
```

The full model gives $F(5, 92) = 32.1747$; the education-only model on the same 98 records gives $F(1, 96) = 47.1956$. Two remarks keep these in proportion. They test different joint restrictions in different models, so the larger value for m_0 is not evidence that m_0 is the better model. And neither statistic verifies the mean function, the error model, or the independence of the rows; a large overall F is compatible with a badly misspecified equation. Nor does either say how a fitted rule would perform on records it has not seen, which is a separate question with a separate calculation (Chapter 13).

3.2 Restrictions in matrix form

The same statistic has a compact matrix form, and the matrix notation is worth one pass because it makes clear what “the same records” is doing in every comparison above. Let $\mathbf{X}$ hold n rows and p coefficient columns including the intercept, with full column rank. Minimizing $(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$ gives the normal equations $\mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}$ and hence $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, and $\hat{\mathbf{V}} = s^2(\mathbf{X}'\mathbf{X})^{-1}$. For q independent linear restrictions $\mathbf{R}\beta = \mathbf{r}$,

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r})}{q s^2} \sim F_{q, n-p}.$$

Choosing the rows of $\mathbf{R}$ that select the two type coefficients, with $\mathbf{r} = \mathbf{0}$, reproduces the adjusted type block test of Section 2.2 without fitting a second model.

```

X <- model.matrix(m2)
beta_hat <- solve(crossprod(X), crossprod(X, d$income_k))
c(columns = ncol(X), rank = qr(X)$rank,
  max_coefficient_difference = max(abs(beta_hat - coef(m2))),
  max_normal_equation_error = max(abs(crossprod(X, residuals(m2)))))

```

	columns	rank
	6.000000e+00	6.000000e+00
max_coefficient_difference	max_normal_equation_error	
	2.057243e-13	5.968559e-13

```

R <- rbind(c(0, 0, 0, 0, 1, 0), c(0, 0, 0, 0, 0, 1))
Rb <- R %*% coef(m2)
drop(t(Rb) %*% solve(R %*% vcov(m2) %*% t(R)) %*% Rb) / nrow(R)

```

```
[1] 0.03189696
```

The model matrix has six columns and rank 6, so the two indicators plus the intercept are not collinear. Solving the normal equations directly reproduces `lm()`'s coefficients to within about 10^{-13} , and the residuals are orthogonal to every column of $\mathbf{X}$ to the same precision. The restriction statistic returns 0.0319, the value the `anova()` comparison gave.

Two practical points follow. `lm()` does not form $(\mathbf{X}'\mathbf{X})^{-1}$; it uses a QR decomposition, which is numerically better behaved. The displayed matrix equation explains what the estimator *is*, not how R computes it (Fox, 2016; Fox & Weisberg, 2019). And because $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto the column space of $\mathbf{X}$, any full-rank recoding of a factor spans the same space and leaves the fitted vector alone — which is Section 1.3's relabeling result stated geometrically. Appendix A carries the derivation.

3.3 A local Normal likelihood and information criteria

Nested F tests compare two models when one contains the other. Information criteria rank models that need not be nested, using a different quantity: the maximized likelihood, penalized for the number of fitted parameters. Chapter 6 built a likelihood once, for 32 successes in 50 binomial trials, by treating the probability of the observed data as a function of the unknown parameter. The linear-model version is the same idea with a continuous density.

Treat the predictor rows as fixed and the outcomes as independent Normal draws with mean $\mathbf{x}_i'\beta$ and SD σ . The log-likelihood, and its value at the maximizing arguments, are

$$\ell(\beta, \sigma) = -n \log \sigma - \frac{n}{2} \log(2\pi) - \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_i' \beta)^2, \quad \ell(\hat{\beta}, \hat{\sigma}_{\text{ML}}) = -\frac{n}{2} \{\log(2\pi) + 1 + \log(RSS/n)\}.$$

Only the final term of the first expression involves β , and it is $-RSS/(2\sigma^2)$. Maximizing over the coefficients therefore *minimizes* the residual sum of squares: least squares and maximum likelihood give the same coefficients under this error model. They do not give the same scale estimate. The maximizing σ is $\sqrt{RSS/n} = 2.5373$, while the residual SD reported by `lm()` is $\sqrt{RSS/(n-p)} = 2.6187$, which divides by the residual degrees of freedom instead. The criteria use the first.

$$AIC = -2 \log \hat{L} + 2k, \quad BIC = -2 \log \hat{L} + k \log n.$$

Here k counts the fitted coefficients **and** the residual variance, so `m2` has $k = 7$, not 6.

```
ll <- logLik(m2)
sigma_ml <- sqrt(mean(residuals(m2)^2))
manual_ll <- sum(dnorm(d$income_k, mean = fitted(m2), sd = sigma_ml,
  log = TRUE))
c(sigma_ML = sigma_ml, residual_SD = sigma(m2),
  logLik = as.numeric(ll), manual = manual_ll, k = attr(ll, "df"),
  AIC = AIC(m2), BIC = BIC(m2))
```

sigma_ML	residual_SD	logLik	manual	k
2.537252	2.618682	-230.301991	-230.301991	7.000000
AIC	BIC			
474.603983	492.698755			

`logLik(m2)` returns -230.3020 , and summing the Normal log-densities of the 98 observed incomes at the fitted means and the ML scale returns the same number. The criteria follow arithmetically: $AIC = 2(230.3020) + 2(7) = 474.6040$ and $BIC = 460.6040 + 7 \log 98 = 492.6988$, with $\log 98 = 4.5850$.

Table 2*In-sample summaries for comparable income models.*

Model	k	R^2	Adjusted R^2	AIC	BIC
m0	3	.3296	.3226	526.5052	534.2601
m1	4	.4996	.4891	499.8328	510.1727
m2	7	.6362	.6164	474.6040	492.6988

Smaller is preferred under either criterion, and both prefer m2 here. The conditions for reading the table are strict and easy to breach. The three models must be fitted to the same records, with the same outcome on the same scale, under the same likelihood convention; Section 4.2 shows what happens when the last of these fails. BIC's heavier penalty, $k \log n$ against AIC's $2k$, means the two can disagree, and a disagreement is information about how strongly the extra parameters are being charged, not a tie to be broken by preference.

Above all, neither criterion is a held-out error. Both are calculated on the same records that fitted the model, and both are comparative rather than absolute: a smaller AIC says one candidate is preferred to another, not that either describes the records well (Akaike, 1974; Schwarz, 1978). Chapter 11 develops likelihood properly, for models where least squares is no longer available; this section needs only the local Normal case.

4 Change the Form of a Predictor or the Scale of the Outcome

4.1 A polynomial term

Chapter 8, Section 4.3, refitted the full model with a squared prestige term as a disclosed sensitivity check on functional form. Here that refit is the worked example, with its block test, its criteria, and the reading a curved term forces.

A squared term is an ordinary extra column, $I(\text{prestige}^2)$, and the model is still linear in its coefficients. What changes is the reading:

$$E(Y \mid x, \dots) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots, \quad \frac{\partial E(Y)}{\partial x} = \beta_1 + 2\beta_2 x, \quad x^* = -\frac{\beta_1}{2\beta_2}.$$

The predictor no longer has a single slope. It has a slope that depends on its own value, and a turning point where that slope is zero.

```
m2q <- lm(income_k ~ education + prestige + I(prestige^2) + women + type,
  data = d)
coef(summary(m2q))[c("prestige", "I(prestige^2)"), ]
```

	Estimate	Std. Error	t value	Pr(> t)
prestige	-0.075446722	0.0900389288	-0.8379345	0.40426298
I(prestige^2)	0.002419616	0.0009322777	2.5953813	0.01101459

```
anova(m2, m2q)
```

Analysis of Variance Table

```
Model 1: income_k ~ education + prestige + women + type
Model 2: income_k ~ education + prestige + I(prestige^2) + women + type
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1     92 630.89
2     91 587.41  1    43.481 6.736 0.01101 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
c(AIC = AIC(m2q), BIC = BIC(m2q),
  turning_point = -coef(m2q)["prestige"] / (2 * coef(m2q)["I(prestige^2)"]))
```

	AIC	BIC
turning_point.prestige	469.60575	490.28549
	15.59064	

```
d$prestige_c <- d$prestige - mean(d$prestige)
m2qc <- lm(income_k ~ education + prestige_c + I(prestige_c^2) + women +
  type, data = d)
c(cor(d$prestige, d$prestige^2), cor(d$prestige_c, d$prestige_c^2),
  max(abs(fitted(m2q) - fitted(m2qc))))
```

```
[1] 9.848978e-01 3.216014e-01 4.618528e-14
```

The quadratic coefficient is 0.0024 with SE 0.0009 and $t = 2.5954$; the one-restriction block test gives $F(1, 91) = 6.7360$ with $p = .0110$; and the criteria move from 474.6040 and 492.6988 to 469.6058 and 490.2855. Adding the term also changes the education coefficient from 0.1164

to -0.0298 , a reminder that coefficients in an additive model are defined relative to the other columns present.

Read the curve by evaluating its slope. With $\beta_1 = -0.0754$ and $\beta_2 = 0.0024$, the fitted slope $-0.0754 + 2(0.0024)x$ is 0.0697 at prestige 30, 0.1665 at 50, and 0.2633 at 70: income rises with prestige throughout, and rises faster at the top. The turning point 15.5906 lies below the smallest observed prestige of 17.3 , so the descending branch of the parabola is entirely outside the records. Reporting a “negative prestige coefficient” from this model would be a misreading of the coordinate -0.0754 , which is the slope at prestige zero, a value no occupation has.

Centering repairs a second coordinate problem. Prestige and its square correlate $.9849$, which inflates both coefficients’ standard errors; centering prestige before squaring lowers that correlation to $.3216$ and changes the linear coefficient to 0.1536 , now the slope at the mean prestige. The two fits have identical fitted values to about 10^{-14} . Centering changes the coordinates in which the curve is described; it does not change the curve.

Finally, keep the status of the exercise straight. This is a disclosed sensitivity check on functional form, declared before the refit, and its result is that a modest curvature is detectable on these records. It is not a search across candidate terms for the smallest p -value, and a p -value read after such a search would not mean what it appears to mean.

4.2 A log-transformed outcome

Changing a predictor’s form keeps the question. Changing the outcome’s scale changes the question, and that difference is the point of this subsection.

$$\log Y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \quad 100(e^{\beta_j} - 1) \text{ percent per unit of } x_j.$$

A coefficient in a log-income model is an additive difference in expected log income. Exponentiating turns it into a multiplicative difference in geometric mean income under that model, and $100(e^{\hat{\beta}} - 1)$ expresses that multiplier as a percentage. The geometric mean is not the arithmetic mean, so this is not automatically a statement about average income in dollars.

```

m_log <- lm(log(income) ~ education + prestige + women + type, data = d)
terms <- c("education", "prestige", "women")
log_ci <- confint(m_log)[terms, ]
data.frame(term = terms, log_coefficient = coef(m_log)[terms],
  percent = 100 * (exp(coef(m_log)[terms]) - 1),
  lower = 100 * (exp(log_ci[, 1]) - 1),
  upper = 100 * (exp(log_ci[, 2]) - 1), row.names = NULL)

```

	term	log_coefficient	percent	lower	upper
1	education	0.007631506	0.7660700	-4.204097	5.994104
2	prestige	0.020748107	2.0964846	1.435058	2.762224
3	women	-0.008465450	-0.8429719	-1.018907	-0.666724

Table 3

Log-model contrasts on a geometric-mean percent scale.

Term	Log coefficient	Percent	Lower	Upper
education	0.0076	0.7661	-4.2041	5.9941
prestige	0.0207	2.0965	1.4351	2.7622
women	-0.0085	-0.8430	-1.0189	-0.6667

One more prestige point is associated with about 2.0965 percent higher geometric mean income at the same education, percentage of women, and type, with interval [1.4351, 2.7622] percent. One more percentage point of women is associated with about -0.8430 percent. The two type coefficients, -0.0137 and 0.0708 on the log scale, are -1.3584 and 7.3341 percent. The intervals are the classical log-scale intervals with both endpoints transformed, which is legitimate because the transformation is monotone; they inherit the log-scale working model and supply no new warrant.

The criteria will not travel across this change. This model's AIC is 8.0910 against `m2`'s 474.6040, and the difference says nothing at all: the two likelihoods are densities of different response variables, so their reference measures differ and their values are not on a common scale. A valid comparison across a transformed outcome needs the Jacobian adjustment that converts one density to the other. The simpler discipline is the one to keep: fix the outcome and the question inside any comparison, and treat a scale change as a new question rather than as data cleaning (Fox & Weisberg, 2019).

4.3 Coefficient scales: standardized coefficients and unique contributions

Two summaries are often asked for once several predictors are in an equation, and both are more limited than their names suggest.

$$\beta_j^* = \hat{\beta}_j \frac{s_{x_j}}{s_y}, \quad \Delta R_j^2 = R_{\text{full}}^2 - R_{\text{without } j}^2.$$

A **standardized coefficient**, or beta weight, rescales a partial slope into outcome-SD units per predictor SD. A **drop-one** R^2 reports how much of the explained variation disappears when one predictor is removed from the full model.

```
coef(m2)[2:4] * sapply(d[c("education", "prestige", "women")], sd) /  
  sd(d$income_k)
```

```
education    prestige      women  
0.07569975  0.56848664 -0.39712218
```

```
summary(m2)$r.squared -  
  summary(lm(income_k ~ prestige + women + type, data = d))$r.squared
```

```
[1] 0.00070774
```

The beta weights are 0.0757, 0.5685, and -0.3971 for education, prestige, and the percentage of women. Education's drop-one R^2 is .0007: removing it from the full model costs essentially nothing in fit, which is the same message the education coefficient's wide interval carried in Chapter 8.

Both quantities are rescalings of an in-sample fit, and neither measures importance. A standardized coefficient depends on the predictor's observed spread in these particular records, so a restricted range shrinks it for reasons that have nothing to do with the relationship. It also differs from the standardization over named profiles that Chapter 8 used, which averages fitted values over a declared distribution rather than dividing by an SD. The drop-one change describes only the *unique* contribution: because predictor information overlaps, the separate drop-one values need not sum to the joint R^2 , and a predictor that duplicates another can show a near-zero value while mattering. Neither number establishes a causal ranking or predicts held-out performance.

5 Robust Uncertainty Under Unequal Spread

5.1 The HC3 covariance

Chapter 7 used a robust standard error once, on the simple regression line, where the education slope's SE moved from 0.1270 to 0.1491. The estimator behind that number is worth stating, because the rest of this chapter turns on what it does and does not change.

The classical covariance $s^2(\mathbf{X}'\mathbf{X})^{-1}$ assumes one common error variance for every row. The heteroskedasticity-consistent family replaces the single s^2 by each row's own squared residual, sandwiched between two copies of the design component:

$$\widehat{\mathbf{V}}_{\text{HC3}} = (\mathbf{X}'\mathbf{X})^{-1} \left\{ \sum_i \mathbf{x}_i \mathbf{x}_i' \frac{e_i^2}{(1 - h_{ii})^2} \right\} (\mathbf{X}'\mathbf{X})^{-1}.$$

The outer factors are the design component; the inner sum is the residual-scale component. Dividing by $(1 - h_{ii})^2$ inflates the contribution of high-leverage rows, which is why HC3 is the usual small-sample recommendation in this family (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980; Zeileis, 2004).

```
V_hc3 <- sandwich::vcovHC(m2, type = "HC3")
cbind(classical = sqrt(diag(vcov(m2))), HC3 = sqrt(diag(V_hc3)))
```

	classical	HC3
(Intercept)	1.957348756	2.774081159
education	0.275226234	0.382332717
prestige	0.035365292	0.038671496
women	0.009662972	0.007856968
typeprof	1.463092959	2.092836846
typewc	0.984774036	1.054843728

```
coef(m2)["education"] + c(-1, 1) * qt(.975, 92) *
  sqrt(V_hc3["education", "education"])
```

```
[1] -0.6429122  0.8757794
```

Table 4*Classical and HC3 standard errors for the full occupation model.*

Coefficient	Estimate	Classical SE	HC3 SE
Intercept	0.4205	1.9573	2.7741
education	0.1164	0.2752	0.3823
prestige	0.1406	0.0354	0.0387
women	−0.0535	0.0097	0.0079
prof	0.3242	1.4631	2.0928
wc	0.2353	0.9848	1.0548

Read the table by column and then by row. The estimate column is unchanged: HC3 replaces a covariance estimate, not a fit, so every coefficient, fitted value, and residual is exactly what it was. The education SE grows from 0.2752 to 0.3823, and its 95% interval on the robust covariance is $[-0.6429, 0.8758]$, wider than the classical one but leading to the same reading of an imprecisely estimated comparison.

The women row is the instructive one: its SE *falls* from 0.0097 to 0.0079. A robust covariance is not a penalty applied to every coefficient. It is a different estimate of the same target, and where the classical common-variance assumption happened to overstate a coefficient’s variability, the robust version is smaller. Reporting only the coefficients whose intervals widened, or switching estimators after seeing which choice gives the preferred conclusion, turns a stated method into a selection.

5.2 A block test with a robust covariance

The restriction statistic of Section 3.2 needs only a coefficient vector and a covariance matrix. Substituting $\widehat{\mathbf{V}}_{\text{HC3}}$ for $\widehat{\mathbf{V}}$ gives an approximate Wald version of the same block test.

```
wald_F <- function(fit, V, which) {
  b <- coef(fit)[which]
  drop(t(b) %*% solve(V[which, which]) %*% b) / length(which)
}
type_cols <- c("typeprof", "typewc")
c(classical = wald_F(m2, vcov(m2), type_cols),
  HC3 = wald_F(m2, V_hc3, type_cols))
```

```
classical      HC3
0.03189696 0.03450534
```

```
pf(wald_F(m2, V_hc3, type_cols), 2, 92, lower.tail = FALSE)
```

```
[1] 0.9660957
```

The type block gives 0.0319 with the classical covariance and 0.0345 with HC3, $p = .9661$ against the classical .9686. Here the substitution changes nothing of substance, which is the ordinary case when the coefficients are far from the restriction.

The reference distribution is where the two differ in kind. The classical statistic has an exact $F_{2,92}$ distribution under the stated normal linear model. The robust version has only an approximate reference, justified as the sample grows, and F with the same degrees of freedom is a convention rather than a derivation. Quoting the robust p -value to several digits, or treating it as the same object as the classical one because both are labelled F , overstates what the calculation delivers. Section 2.2's classical test and this one answer the same question with different warrants.

5.3 What HC3 does not repair

HC3 has one job. Under a maintained linear conditional mean and independent observations, it estimates the coefficient covariance without assuming a common error variance. That is all.

It does not repair curvature: if the mean function is wrong, the coefficients estimate a different quantity and a robust covariance describes that different quantity's variability more honestly, without making it the quantity of interest. It does not repair dependence: rows that are not independent break the estimator's own derivation, and Section 7 shows a case where a robust row-level SE and a naive one agree closely while both understate the uncertainty by nearly half. It does not repair measurement error, selection, or missing outcomes, which change what the records represent. It does not supply causal identification, which comes from design or from an argument about the conditioning set. And it does not create joint support: the type contrast's robust SE of 1.1827 is still the uncertainty of a comparison the records barely observe.

The practical response is to say which estimator was used and why, and to keep the answer's limits attached to it. Chapter 8's advice holds here: report the named comparison, its form and influence sensitivities, and the covariance calculation, rather than treating any one of them as a pass/fail certificate.

6 Resample the Records

6.1 Why resample, and which scheme

Every uncertainty statement so far came from a formula: a coefficient covariance matrix, classical or robust, propagated through a linear combination. Resampling offers a different approximation. Refit the model to many resamples of the records, and read the spread of the refitted estimates directly (Davison & Hinkley, 1997; Efron, 1979; Efron & Tibshirani, 1993). Where the formula and the empirical resampling distribution disagree, the disagreement is itself informative.

The choice of what to resample is an assumption, not a default.

- **Case resampling** draws whole records with replacement and refits. It conditions on nothing and inherits whatever unequal spread the rows contain, but it requires the resampling unit to match the sampling unit, with units independent and exchangeable.
- **Residual resampling** holds the design fixed, resamples rescaled residuals, and adds them back to the fitted values. It assumes identically distributed errors — the assumption HC3 exists to relax.
- **The wild bootstrap** also holds the design fixed but multiplies each rescaled residual by an independent mean-zero, unit-variance draw, so every record keeps its own error scale. It is the residual scheme repaired for unequal variance.

This chapter uses case resampling, for two stated reasons. The occupation predictors were not designed: no one assigned education or prestige to an occupation, so a joint-sampling view of rows is the natural one. And the residual spread differs visibly by type, so the identically-distributed errors behind residual resampling are not available (Freedman, 1981). Appendix B carries out the other two schemes on the same targets for comparison.

6.2 Case resampling for the full model

Resample the 98 rows with replacement, refit `m2`, and record three targets: the education slope, the professional-minus-white-collar contrast, and the professional coefficient. The seed and the number of resamples B belong in the record, so both appear in the code; B counts bootstrap resamples here, as it counted simulation repetitions in Chapter 5.

```

set.seed(90909)
B <- 2000
coef_names <- names(coef(m2))
draws <- matrix(NA_real_, B, 3,
  dimnames = list(NULL, c("education", "prof_minus_wc", "prof")))
for (b in seq_len(B)) {
  rows <- sample.int(nrow(d), replace = TRUE)
  fit_b <- lm(income_k ~ education + prestige + women + type,
    data = d[rows, ])
  b_hat <- coef(fit_b)
  if (all(coef_names %in% names(b_hat)) && !anyNA(b_hat[coef_names])) {
    draws[b, ] <- c(b_hat["education"], b_hat["typeprof"] - b_hat["typewc"],
      b_hat["typeprof"])
  }
}
colSums(is.finite(draws))

```

education	prof_minus_wc	prof
2000	2000	2000

```

rbind(estimate = c(coef(m2)["education"], C_hat, coef(m2)["typeprof"]),
  classical_SE = c(sqrt(V["education", "education"]), sqrt(C_var),
    sqrt(V["typeprof", "typeprof"])),
  HC3_SE = c(sqrt(V_hc3["education", "education"]),
    sqrt(drop(c_vec %*% V_hc3 %*% c_vec)),
    sqrt(V_hc3["typeprof", "typeprof"])),
  bootstrap_SE = apply(draws, 2, sd, na.rm = TRUE),
  lower = apply(draws, 2, quantile, .025, na.rm = TRUE, names = FALSE),
  upper = apply(draws, 2, quantile, .975, na.rm = TRUE, names = FALSE))

```

	education		typeprof
estimate	0.1164336	0.08886278	0.3241934
classical_SE	0.2752262	1.07999307	1.4630930
HC3_SE	0.3823327	1.18271129	2.0928368
bootstrap_SE	0.3698895	1.11869855	1.9925695
lower	-0.6829568	-1.96126245	-3.1415854
upper	0.7439809	2.43128069	4.5443286

The printed matrix carries one blank column name: the middle target is a contrast rather than a coefficient, so `C_hat` arrives without a name. Its column is the professional-minus-white-collar comparison of Section 2.1, and the table below labels all three targets explicitly.

The guard inside the loop matters. A resample could omit an entire occupational type, and the refit would then drop a coefficient; such a draw must be recorded as missing rather than silently misaligned. Here all 2,000 resamples contained all three types, so all 2,000 refits are usable, and that count is part of the result rather than an assumption.

Table 5

Model-based, HC3, and case-bootstrap uncertainty for three targets in the full occupation model (seed 90909, $B = 2000$).

Target	Estimate	Classical SE	HC3 SE	Bootstrap SE	Percentile 2.5%	Percentile 97.5%
Education slope	0.1164	0.2752	0.3823	0.3699	−0.6830	0.7440
prof — wc	0.0889	1.0800	1.1827	1.1187	−1.9613	2.4313
prof coefficient	0.3242	1.4631	2.0928	1.9926	−3.1416	4.5443

Read the table one row at a time. The education slope’s bootstrap SE of 0.3699 sits between the classical 0.2752 and the HC3 0.3823, with percentile interval [−0.6830, 0.7440]: three calculations, one conclusion that this comparison is imprecise. The type contrast’s bootstrap SE of 1.1187 exceeds the classical 1.0800, with percentile interval [−1.9613, 2.4313]. The professional coefficient shows the largest movement, from a classical 1.4631 to 1.9926, close to the HC3 2.0928; with only 31 professional occupations, a resample’s composition matters, and the formula that assumes a common error variance is the most optimistic of the three.

Agreement between HC3 and the bootstrap is not independent confirmation. Both are sensitivity procedures applied to the same 98 records, and both inherit the same limitations of those records; agreement means the two approximations behave similarly here, not that the answer has been replicated. Monte Carlo variation also means that the last decimal of a bootstrap summary carries no weight: a different seed would move it, which is exactly why the seed and $B = 2000$ are reported.

6.3 Which interval, and what resampling does not repair

Two intervals come directly from the resampling distribution. The **percentile interval** takes its 2.5th and 97.5th quantiles, as in the table above. The **normal-approximation interval** takes the original estimate plus and minus 1.96 bootstrap standard errors. The two agree when the resampling distribution is roughly symmetric; here the three distributions have skewness −0.3891, 0.2635, and 0.3785, mild enough that the endpoints agree closely. For a strongly skewed or biased resampling distribution, **bootstrap- t** intervals, which resample a studentized quantity, and **BCa** intervals, which correct for bias and for skewness-driven acceleration, are the higher-order alternatives; Appendix B defines both in words.

What resampling cannot do is more important than which interval to choose. Drawing occupations with replacement treats them as independent and exchangeable draws, which they are not: they share one historical labour market, and their measurements are aggregates from different years. Resampling does not repair the thin cross-type support, correct the measurement problems Chapter 8 documented, or convert an observational comparison into a causal one. And if a design contained clusters, strata, or weights, the resampling unit would have to follow the design rather than the rows — which is the subject of the next section.

7 Dependence in the Design: Project STAR

Every calculation so far has used one set of 98 observational records, and every qualification has been about what those records cannot support. This section changes the design. Project STAR randomized kindergarten pupils to class types within their schools, which supplies a warrant no amount of adjustment can manufacture — and a dependence structure that the uncertainty calculation must follow.

7.1 A different design supplies a different warrant

The question is specific:

Among eligible kindergarten pupils in the participating Tennessee schools, what was the mean reading-score difference under assignment to a small rather than a regular class?

The outcome is `readk`, an end-of-kindergarten reading scale score. The field `stark` records the kindergarten class-type condition. Reading its small-versus-regular comparison as an assignment comparison rests on the documented experimental protocol, not on the column name or on the `lm()` call: random assignment occurred **within** participating schools (Finn & Achilles, 1990; Mosteller, 1995). The regular-with-aide arm is outside this declared two-arm comparison and is excluded by design, not dropped for convenience.

Three limits belong with the question from the start. Class **type** is not a measured headcount, so this analysis estimates no dose response in pupils per room. The distributed field does not independently verify each pupil’s original randomized assignment; an assignment-effect reading additionally requires linking the recorded type to the assignment record and checking switches and implementation deviations, which this teaching reanalysis does not supply. And the scientific target is the pupil-weighted mean assignment effect for the eligible cohort in participating schools, while a common coefficient from a complete-record model is an estimator that needs further links — a common effect across schools, or an explicit weighting argument — to reach that target.

The data are distributed as `AER::STAR`, accompanying *Applied Econometrics with R* and following the version used by Stock and Watson (Kleiber & Zeileis, 2008; Stock & Watson, 2007). They are loaded from the package rather than redistributed here, and the analysis is retrospective rather than preregistered.

7.2 Separate eligibility from missingness

Two different reasons for a record's absence from an analysis are easy to merge and must not be. Some distributed rows belong to pupils who entered the study after kindergarten: they have no kindergarten condition at all, and their absence is **structural ineligibility**. Other rows belong to eligible kindergarten pupils whose reading score or covariates were not recorded: that is **missingness**, and it can distort a comparison.

Only variables a calculation actually uses may enter its complete-record rule. In particular `mathk` must not exclude an otherwise usable reading record.

```
data("STAR", package = "AER")
star_vars <- c("stark", "readk", "lunchk", "gender", "ethnicity",
  "experiencek", "schoolidk")
star_k_recorded <- !is.na(STAR$stark)
star_two_arm <- star_k_recorded & STAR$stark %in% c("regular", "small")
star_complete <- complete.cases(STAR[, star_vars])
star_df <- droplevels(STAR[star_two_arm & star_complete, star_vars])
star_df$stark <- relevel(star_df$stark, ref = "regular")
star_observed <- droplevels(STAR[star_two_arm & !is.na(STAR$readk), c(
  "stark", "readk", "schoolidk")])
star_observed$stark <- relevel(star_observed$stark, ref = "regular")
c(observed_reading = nrow(star_observed), complete_covariates = nrow(
  star_df))
```

observed_reading	complete_covariates
3745	3735

Table 6
STAR eligibility and reading-analysis completeness.

Stage	<i>n</i>
Distributed records	11,598
Kindergarten condition recorded	6,325
Eligible small or regular	4,094
Eligible, reading observed	3,745
Complete primary records	3,735
Primary regular	2,001
Primary small	1,734
Primary schools	79

Of 11,598 distributed records, 5,273 have no kindergarten condition; they are later entrants, not 5,273 failed measurements. The remaining 6,325 have a recorded kindergarten arm, and across all three arms 536 lack a reading score. Restricting to the two declared arms leaves 4,094 eligible records, of which 349 lack reading — 188 in the regular arm and 161 in the small arm — leaving 3,745 with an observed reading score, 2,006 regular and 1,739 small.

Similar missing percentages in the two arms are reassuring but prove nothing: the absent scores could differ systematically within either arm, and no observed value can reveal that. Complete-record inference here requires an assumption that, given the declared covariates and within arm and school, observation does not depend on the reading score that would have been recorded. Stating the assumption is not verifying it.

Among the observed-reading records across all arms, 38 lack at least one of lunch status, gender, ethnicity, and teacher experience; within the two focal arms, ten do. The primary common-covariate sample therefore has 3,735 pupils, 2,001 regular and 1,734 small, in 79 schools. The observed-reading sample keeps all 3,745 focal records and is used as a declared sensitivity, not as an alternative headline.

7.3 The binary slope recovers the observed arm means

Section 1.1’s identity applies unchanged. With $D_i = 1$ for the small arm and 0 for regular,

$$\widehat{Y}_i = \widehat{\alpha} + \widehat{\delta}D_i, \quad \widehat{\alpha} = \bar{Y}_{D=0}, \quad \widehat{\delta} = \bar{Y}_{D=1} - \bar{Y}_{D=0} = 440.5474 - 434.7323 = 5.8151.$$

```

star_observed_unadjusted <- lm(readk ~ stark, data = star_observed)
star_arm_means <- tapply(star_observed$readk, star_observed$stark, mean)
c(regular_mean = unname(star_arm_means["regular"]),
  small_mean = unname(star_arm_means["small"]),
  small_minus_regular = unname(star_arm_means["small"] -
    star_arm_means["regular"]),
  regression_difference = unname(coef(
    star_observed_unadjusted)["starksmall"]))

```

	regular_mean	small_mean	small_minus_regular
	434.732303	440.547441	5.815138
regression_difference			5.815138

On these 3,745 records the difference is 5.8151 reading points, and the regression slope reproduces it exactly, as Chapter 7's binary predictor did on 24 pupils. The identity is algebraic and requires no randomization; what randomization adds is the reason to read the difference as an assignment comparison.

Two cautions attach to the number itself. Under independent common-variance rows, the slope's t test equals the pooled two-sample test, here $t = 5.6039$; that row-level reference ignores the fact that pupils sit in schools, which the next subsection addresses. And raw arm means weight each school by its own arm-specific pupil count, so the pooled difference equals the pupil-weighted cohort target only under common assignment probabilities and comparable observation patterns, or after explicit standardization to target school weights.

7.4 Compare two specifications on identical primary rows

The adjusted model adds free-lunch status, gender, ethnicity, teacher experience, and school indicators. School indicators respect the assignment blocks; the remaining covariates are the declared adjustment specification, not a substitute for randomization.

```

star_unadjusted <- lm(readk ~ stark, data = star_df)
star_primary <- lm(readk ~ stark + lunchk + gender + ethnicity +
  experiencek + schoolidk, data = star_df)
star_sensitivity <- lm(readk ~ stark + schoolidk, data = star_observed)
c(unadjusted = nobs(star_unadjusted), adjusted = nobs(star_primary),
  observed_reading = nobs(star_sensitivity))

```

unadjusted	adjusted	observed_reading
3735	3735	3745

On the common 3,735 primary rows, the unadjusted difference is 5.8955 and the adjusted coefficient is 6.6575. Adjustment can improve precision when the added variables are prognostic for reading, and school indicators match the blocks in which assignment happened. It is not needed to remove baseline confounding that a successfully implemented randomization already prevents, and it does not automatically preserve a causal target: timing, selection, and the fitted comparison still need scrutiny.

The earlier 5.8151 comes from 3,745 rows and must not serve as the same-row unadjusted baseline for 6.6575. Comparing 5.8151 with 6.6575 confounds two changes — a change of records and a change of specification — and the whole discipline of this chapter is to keep those apart. The same-row pair to compare is 5.8955 against 6.6575.

Similar unadjusted and adjusted coefficients are consistent with successful randomization. They are not proof that implementation, missingness, or weighting problems are absent, and a common coefficient still carries its own implicit weights if the effect varies across schools or pupil profiles.

7.5 Cluster uncertainty follows the design

Pupils are the outcome units. Schools are the randomization blocks and the approximately independent units for this covariance calculation; the schools themselves were not assigned to arms. Pupils sharing a school — and, within it, a classroom — may have correlated outcomes, and a covariance estimate that ignores this understates the uncertainty of an arm comparison.

For school g , form its score $\mathbf{s}_g = \mathbf{X}'_g \mathbf{e}_g$ by summing the residual contributions **within** the school before taking an outer product. The CR1/HC1 cluster-robust covariance is then

$$\widehat{\mathbf{V}}_{\text{CR1}} = c_G (\mathbf{X}' \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{s}_g \mathbf{s}'_g \right) (\mathbf{X}' \mathbf{X})^{-1}, \quad \mathbf{s}_g = \mathbf{X}'_g \mathbf{e}_g, \quad c_G = \frac{G}{G-1} \frac{n-1}{n-p}.$$

The within-school cross-products are exactly what distinguishes this from treating every pupil as independent, and c_G is the finite-cluster correction. With $G = 79$ schools, the interval below uses a t_{78} critical value, 1.9908 (Zeileis et al., 2020).

```
star_V <- sandwich::vcovCL(star_primary, cluster = star_df$schoolidk,
  type = "HC1", cadjust = TRUE)
star_b <- unname(coef(star_primary)["starksmall"])
star_se <- sqrt(star_V["starksmall", "starksmall"])
star_G <- nlevels(star_df$schoolidk)
c(estimate = star_b, SE = star_se)
```

```
estimate      SE
6.657510 1.697409
```

```
star_b + c(-1, 1) * qt(.975, star_G - 1) * star_se
```

```
[1] 3.278229 10.036791
```

```
c(row_level = sqrt(vcov(star_observed_unadjusted)["starksmall", "starksmall"]),
  HC3 = sqrt(sandwich::vcovHC(star_observed_unadjusted,
    type = "HC3")["starksmall", "starksmall"]),
  clustered = sqrt(sandwich::vcovCL(star_observed_unadjusted,
    cluster = star_observed$schoolidk, type = "HC1",
    cadjust = TRUE)["starksmall", "starksmall"]))
```

```
row_level      HC3 clustered
1.037686 1.041615 1.847617
```

Table 7

Class-type comparisons and school-clustered intervals.

Model	n	Estimate	SE	Interval
Observed: unadjusted	3,745	5.8151	1.8476	[2.1368, 9.4935]
Primary: unadjusted	3,735	5.8955	1.8491	[2.2143, 9.5767]
Primary: adjusted	3,735	6.6575	1.6974	[3.2782, 10.0368]
Observed: school-adjusted	3,745	6.6273	1.7797	[3.0842, 10.1703]

The table holds four distinct calculations, each with its own records and its own specification; each row's interval uses the school identifiers from that model's own rows. Changing the model

or its records is a different act from changing the covariance calculation, and the table keeps them separable.

The three standard errors printed at the end of the chunk make Section 5's point concrete. For the same 3,745-record unadjusted comparison, treating pupils as independent rows gives SE 1.0377; HC3 gives 1.0416, barely different; school clustering gives 1.8476, nearly twice as large. Unequal spread across pupils was not this design's problem. Dependence within schools was, and no row-level robust covariance addresses it. The adjusted primary model shows the same pattern: its row-level SE is 0.9161 and its HC3 SE 0.9509, against the clustered 1.6974.

These are cluster-asymptotic approximations with 79 clusters, not exact randomization inference and not a multilevel model. A residual plot cannot choose the independence unit — the design does. And a count of 79 clusters does not certify the approximation: the balance of information across clusters and the influence of particular schools also matter.

7.6 Read the bounded answer, then evaluate its support

A chapter-authored report can now state:

On 3,735 complete two-arm kindergarten records, the common adjusted small-versus-regular class-type coefficient was 6.6575 reading points, with approximate school-clustered interval [3.2782, 10.0368]. Its causal mean-effect interpretation relies on documented within-school randomization, a verified link from recorded class type to original assignment, suitable complete-record selection, independent-school uncertainty, and a common-effect or target-matching weighting argument. It concerns assigned class type, one kindergarten reading outcome, and participating Tennessee schools, not realized headcount or a portable policy dose response.

Read the report's functions before judging it, applying Chapter 1's argument components as Chapters 7 and 8 did. The sample audit and the model code carry **STAT-E** content; the design and covariance reasoning supplies **STAT-J**; the estimate and its clustered interval supply **STAT-I**; and the first sentence is the bounded **STAT-C** answer. **SUB-E** situates the pupils, the test, the cohort, and the school context; **SUB-J** adds the assignment record together with the selection and weighting qualifications; **SUB-I** reads the coefficient as assignment-related only under that support; and **SUB-C** bounds the substantive claim. These are linked functions within one argument, not two arguments.

Then separate what is *stated* from what is *demonstrated*. The report states the selection and weighting conditions; stating them does not establish that they hold. The randomization protocol is documented in the study's own publications, not shown by this coefficient table. The link from the recorded class-type field to the original assignment is asserted here and not verified. Where a necessary link cannot be assessed, the honest move is to report the limitation rather than to invent it or to quietly substitute a different target.

The design literature documents implementation and interpretation issues that a single coefficient cannot settle (Chetty et al., 2011; Hanushek, 1999; Krueger, 1999; Nye et al., 2000). Nor does this result transport by itself to nonparticipating schools, another period, another outcome, or another education system. Those limits belong inside the answer, not in a list appended after an unrestricted claim — and a crossed significance boundary is not what makes a result (Gelman et al., 2020; Gelman & Stern, 2006).

Chapter 14 returns to dependence as a design fact and reuses this calculation as its example. What a design supplies that adjustment cannot is the subject there; what this section supplies is the discipline of letting the covariance calculation follow the unit at which the design acted.

8 Reading Bridge

A factor enters a regression as indicators, and its fitted means, contrasts, and omnibus test are the analysis of variance written as a model; a contrast’s uncertainty needs the covariance between coefficients, not the sum of two standard errors. Nested models compare on the same records and the same outcome by an extra-sum-of-squares F , and information criteria rank them by penalized in-sample fit, never by held-out error. A squared predictor gives a comparison that depends on the predictor’s value; a logged outcome changes the question to a geometric-mean scale, and its criteria no longer speak to the original one. HC3, the case bootstrap, and a clustered covariance each change the uncertainty calculation for a stated reason and leave the fitted coefficients alone; none repairs curvature, missing outcomes, weighting, or the absence of a design. Project STAR showed the uncertainty following the design: schools, not pupils, were the approximately independent units, and the difference was nearly a factor of two.

Chapter 10 asks a different question of the same equations. Every comparison in this chapter was the same at every value of the other predictors — that is what “additive” means, and Section 2.3’s common-slope check was the first test of it. Chapter 10 lets a comparison itself vary with another predictor, and develops how to read, probe, and report such an interaction.

Key terms

Use this short reference to revisit the concepts introduced here. The Part One introduction gives the broader reference and the principal home of each concept; indicator coding, partial slopes, coefficient covariance, and HC3 are explained in Chapters 7 and 8.

Table 8
Selected concepts for the regression toolbox.

Term and common alternatives	Meaning
Nested-model F test (extra sum of squares; block test)	Compares a reduced model with a fuller one fitted to the same records and outcome: the RSS reduction per added coefficient, divided by the fuller model's residual mean square, referred to F with the number of restrictions and the fuller model's residual df. It tests the added block jointly, not each coefficient separately.
Information criteria (AIC; BIC)	Penalized in-sample fit: $-2 \log \hat{L}$ plus $2k$ (AIC) or $k \log n$ (BIC), with k counting the coefficients and the residual variance. Smaller is preferred among models fitted to the same records and outcome scale; neither is held-out error.
Polynomial term (quadratic term; curvature)	A power of a predictor added as a further column so that the fitted mean can curve. The predictor's comparison then depends on its value; centering before squaring reduces collinearity without changing the fitted curve.
Log-transformed outcome (geometric-mean scale)	Fitting the logarithm of the outcome, so that a coefficient is an additive difference in expected log outcome and $100(e^\beta - 1)$ percent is a multiplicative difference in geometric means. A changed outcome scale is a changed question, and its criteria are not comparable with the original scale's.
Standardized coefficient (beta weight)	A partial slope rescaled by the predictor and outcome SDs, in outcome-SD units per predictor SD, with the other predictors held fixed. It differs from standardization over named profiles and does not measure causal importance.
Bootstrap (case resampling; percentile interval)	Refitting the model to many resamples of the records drawn with replacement and using the spread of the refitted estimates as an SE, or their middle 95% as a percentile interval. It treats the resampled units as independent and exchangeable; it does not repair dependence, thin support, measurement error, or an observational design.
Cluster-robust standard error (school-clustered covariance)	A coefficient covariance that sums residual contributions within clusters, such as schools, before combining them, allowing dependence within clusters under an approximation that treats clusters as independent. It changes uncertainty, not the coefficient, and does not repair missing outcomes or weighting.

Further reading

- The OpenIntro multiple-predictor and model-selection tutorial gives concrete indicator interpretations and criterion-based reading. We use its equation-first device without adopting automatic stepwise selection.
- The ISLR regression lab and the ISLR resampling lab connect fitting, factors, criteria, and resampling in one workflow (James et al., 2021).
- Fox (Fox, 2016) and Kutner and colleagues (Kutner et al., 2005) develop the linear-model algebra, the restriction statistic, and the diagnostics behind this chapter’s matrix section.
- Efron and Tibshirani (Efron & Tibshirani, 1993) and Davison and Hinkley (Davison & Hinkley, 1997) are the standard bootstrap references, including the bootstrap- t and BCa intervals named in Section 6.3.
- The sandwich documentation specifies the HC and CR covariance calls used here (Zeileis et al., 2020).

Exercises

Use the supplied numbers; rounding within 0.02 is acceptable unless a different precision is requested. These prompts do not require the appendices.

Exercise 1: Read indicator coefficients (Foundational)

A group-only regression of income (thousand dollars) on occupation type has observed group means 5.3741 (blue-collar, 44 records), 10.5595 (professional, 31), and 5.0523 (white-collar, 23). (1) With blue-collar as reference, give the intercept and the two indicator coefficients. (2) Give the intercept and both coefficients with white-collar as reference. (3) Explain why the fitted values are the same under both codings. (4) The full additive model’s professional coefficient is 0.3242. Explain why it differs from the professional coefficient in (1) and why neither is “the effect of occupational type”.

Exercise 2: A contrast needs covariance (Foundational)

In the full additive model the professional and white-collar coefficients are 0.3242 and 0.2353, with estimated variances 2.1406 and 0.9698 and covariance 0.9720; the residual df are 92 and $t_{.975,92} = 1.9861$. (1) Estimate the professional-minus-white-collar contrast and its variance. (2) Give its SE and 95% interval. (3) A student adds the two SEs (1.4631 and 0.9848) instead. State what that number is and why it is wrong. (4) Explain what the interval does and does not say about the two types.

Exercise 3: A nested comparison from residual sums of squares (Foundational)

On the same 98 records, the model with education and prestige has RSS 867.6537 on 95 df, and the full model adding the percentage of women and the two type indicators has RSS 630.8897 on 92 df. The 5% critical value of $F(3, 92)$ is 2.7036. (1) Calculate the F statistic for the added block. (2) State the decision at 5% and what it does and does not say about the three added columns. (3) Explain why the sequential sums of squares from a single `anova()` call on the full model would not answer this question.

Exercise 4: Information criteria by hand (Foundational)

The full model's maximized log-likelihood is -230.3020 on 98 records, with seven fitted parameters (six coefficients and the residual variance), and $\log 98 = 4.584967$. (1) Calculate AIC and BIC. (2) The model with education and prestige only has AIC 499.8328. Which model do the criteria favour, and on what condition is that comparison meaningful? (3) A log-income version of the full model has AIC 8.0910. Explain why this number cannot be compared with your answer to (1).

Exercise 5: Percent and curvature (Integrative)

- (1) In a log-income model the prestige coefficient is 0.0207. Give the percent difference in geometric-mean income per prestige point and say what “geometric mean” adds to the reading. (2) In an income model with a squared prestige term, the prestige coefficients are -0.0754 (linear) and 0.0024 (squared); prestige ranges from 17.3 to 87.2. Give the fitted slope at prestige 30, 50, and 70 and locate the turning point. (3) State in one sentence what the curve claims about income and prestige over the observed range, and one reason that claim is a sensitivity check rather than a discovery.

Exercise 6: Choose the uncertainty calculation (Integrative)

In Project STAR the small-minus-regular kindergarten reading difference on 3,745 observed records is 5.8151 points. Three standard errors are available: 1.0377 treating pupils as independent, 1.0416 with HC3, and 1.8476 clustering pupils within their 79 schools; use $t_{.975} = 1.9606$ for the first two and 1.9908 for the third. (1) Give the interval under the first and the third calculation. (2) Say which one a report should use and why, naming the unit at which assignment happened. (3) Name two problems that none of the three calculations repairs.

Appendix A: Matrix Geometry, Restrictions, and Sandwich Constructions

This appendix derives the results that Sections 3.2 and 5.1 used. It is reference material: the exercises do not require it, and the main route is complete without it. Complete R reproductions of every construction below belong to the Computer Lab tasks on normal equations, on manual HC3, and on the joint Wald helper.

Projection and the normal equations

Let $\mathbf{X}$ be $n \times p$ with full column rank, so $\mathbf{X}'\mathbf{X}$ is invertible. The least-squares criterion is $S(\beta) = (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta)$. Expanding gives $S = \mathbf{y}'\mathbf{y} - 2\beta'\mathbf{X}'\mathbf{y} + \beta'\mathbf{X}'\mathbf{X}\beta$, and differentiating with respect to β yields the normal equations

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}, \quad \text{equivalently} \quad \mathbf{X}'(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathbf{0}.$$

The second form is the geometry: the residual vector is orthogonal to every column of $\mathbf{X}$. Hence $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$, and $\mathbf{H}$ is symmetric and idempotent — the orthogonal projection onto the column space of $\mathbf{X}$. Its trace is p , which is why the leverages h_{ii} sum to the number of fitted coefficients; for `m2` they sum to 6.

Two consequences used earlier follow immediately. Any full-rank recoding $\mathbf{XA}$, with $\mathbf{A}$ invertible, spans the same column space, so $\mathbf{H}$ and $\hat{\mathbf{y}}$ are unchanged: releveling a factor or centering a predictor cannot alter a single fitted value. Adding a genuinely new column enlarges the space and can change every fitted value. And under the classical error model, $\hat{\mathbf{V}} = s^2(\mathbf{X}'\mathbf{X})^{-1}$ with $s^2 = RSS/(n-p)$, which is where every classical standard error in this chapter came from. Section 3.2's check confirmed both the coefficients and the orthogonality to about 10^{-13} on the occupation model.

General linear restrictions

Let $\mathbf{R}$ be $q \times p$ of rank q and consider $\mathbf{R}\beta = \mathbf{r}$. Since $\hat{\beta}$ has covariance $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ under the classical model, the vector $\mathbf{R}\hat{\beta} - \mathbf{r}$ has covariance $\sigma^2\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'$. Standardizing by that covariance and dividing by q gives

$$F = \frac{(\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r})}{q s^2},$$

which has an exact $F_{q, n-p}$ reference under normal errors, because the numerator and denominator quadratic forms are independent chi-squares. This is the same number the extra-sum-of-squares formula produces: fitting the restricted model and differencing residual sums of squares is an algebraically equivalent route. Taking the two rows of $\mathbf{R}$ that select `typeprof` and `typewc`, with $\mathbf{r} = \mathbf{0}$, returned 0.0319 in Section 3.2 — the adjusted type block test, obtained without fitting the reduced model.

The equivalence with the extra-sum-of-squares route is worth seeing once. Minimizing $S(\beta)$ subject to $\mathbf{R}\beta = \mathbf{r}$ with a Lagrange multiplier gives the restricted estimator $\tilde{\beta} = \hat{\beta} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'[\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1}(\mathbf{R}\hat{\beta} - \mathbf{r})$, a correction applied only in the directions the restriction touches. Substituting it into the criterion shows that the increase in residual sum of squares is $RSS_R - RSS_F = (\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r})$, which is exactly the numerator above. Fitting the reduced model and differencing, or evaluating the quadratic form in the unrestricted fit, are two routes to one number.

Replacing $s^2(\mathbf{X}'\mathbf{X})^{-1}$ by a robust covariance in the middle of this expression gives the Wald statistic of Section 5.2. The algebra survives the substitution; the exact reference distribution does not.

The HC3 sandwich

Write the coefficient error as $\hat{\beta} - \beta = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\varepsilon$. Under independent rows with possibly different variances σ_i^2 , its exact covariance is

$$(\mathbf{X}'\mathbf{X})^{-1} \left(\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right) (\mathbf{X}'\mathbf{X})^{-1},$$

a *sandwich* with the design component as bread and the error-scale component as filling. The classical estimator substitutes a common s^2 , which collapses the filling to $s^2 \mathbf{X}'\mathbf{X}$. The HC family substitutes each row's own squared residual instead. Because $E(e_i^2) = \sigma_i^2(1 - h_{ii})$ under the classical model, the raw residual understates the error at high-leverage rows; HC3 corrects by dividing by $(1 - h_{ii})^2$, which is deliberately slightly conservative and performs well in small samples (Long & Ervin, 2000; MacKinnon & White, 1985).

Building the estimator directly — inverting $\mathbf{X}'\mathbf{X}$ for the bread and forming the cross-product of the rows of $\mathbf{X}$ scaled by $e_i/(1 - h_{ii})$ for the filling — reproduces `sandwich::vcovHC(m2, type = "HC3")` to within about 10^{-12} on the occupation model, which is floating-point agreement.

The CR1 cluster sandwich

Cluster robustness changes the filling, not the bread. Partition the rows into G clusters and let $\mathbf{X}_g$ and $\mathbf{e}_g$ hold cluster g 's rows and residuals. The cluster score $\mathbf{s}_g = \mathbf{X}_g' \mathbf{e}_g$ sums the within-cluster contributions *before* the outer product is taken, so the estimator retains the within-cluster residual correlations that the independent-row filling discards:

$$\widehat{\mathbf{V}}_{\text{CR1}} = c_G (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{s}_g \mathbf{s}_g' \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad c_G = \frac{G}{G-1} \frac{n-1}{n-p}.$$

If every cluster held one row, the filling would reduce to the HC0 form, which is why a row-level robust SE is the special case that assumes away the dependence. The asymptotics are in G , not in n : 3,745 pupils in 79 schools supply 79 independent pieces of information for this covariance, which is why the clustered SE of 1.8476 is so much larger than the row-level 1.0377 and why a t_{G-1} reference is used. Constructing the score matrix by summing $\mathbf{X} \odot \mathbf{e}$ within school, applying $c_G = 1.0131$, and sandwiching it reproduces `sandwich::vcovCL(..., type = "HC1", cadjust = TRUE)` to within about 10^{-13} on the observed-reading STAR model.

Appendix B: Bootstrap Schemes and Intervals

Section 6 used case resampling and said why. This appendix runs the other two schemes on the same model, the same three targets, and the same seed, and defines the two higher-order intervals named there. The lab tasks on resampling variants and on manual covariance reproduction carry the full implementations.

Residual and wild resampling

Both schemes hold the design matrix fixed and rebuild outcomes from the fitted values. Residual resampling draws leverage-rescaled residuals $e_i/\sqrt{1-h_{ii}}$ with replacement and adds them to the fitted values, which assumes the rescaled residuals are exchangeable across rows. The wild bootstrap keeps each row's own rescaled residual and multiplies it by an independent mean-zero, unit-variance draw — here ± 1 with equal probability — so each row retains its own error scale.

```
set.seed(90909)
X <- model.matrix(m2); e <- residuals(m2); h <- hatvalues(m2)
fit_draw <- function(y) {
  b <- coef(lm(y ~ X - 1))
  c(b["Xeducation"], b["Xtypeprof"] - b["Xtypewc"], b["Xtypeprof"])
}
residual_draws <- t(replicate(B, fit_draw(fitted(m2) +
  sample(e / sqrt(1 - h), replace = TRUE))))
wild_draws <- t(replicate(B, fit_draw(fitted(m2) +
  e / sqrt(1 - h) * sample(c(-1, 1), nrow(d), replace = TRUE))))
rbind(case = apply(draws, 2, sd, na.rm = TRUE),
  residual = apply(residual_draws, 2, sd), wild = apply(wild_draws, 2, sd))
```

	education	prof_minus_wc	prof
case	0.3698895	1.118699	1.992570
residual	0.2819546	1.108103	1.486120
wild	0.3673166	1.115091	1.976719

For the education slope the three schemes give 0.3699 (case), 0.2820 (residual), and 0.3673 (wild). The residual scheme is the outlier, and its value sits close to the classical 0.2752 — which is what one should expect, since both assume one common error distribution. The wild bootstrap, which does not, lands near the case result and near HC3's 0.3823. For the professional coefficient the pattern repeats more strongly: 1.9926, 1.4861, and 1.9767. When the schemes disagree, the disagreement is about the error model, not about the data, and it is a reason to say which scheme was used.

Percentile, normal, and higher-order intervals

Four interval definitions are worth distinguishing in words.

- The **percentile interval** takes the 2.5th and 97.5th quantiles of the resampling distribution of the estimate.

- The **normal-approximation interval** takes the original estimate plus and minus 1.96 bootstrap standard errors. It uses the resampling distribution only for its spread, so it is insensitive to skewness.
- The **bootstrap- t interval** resamples a studentized quantity, $(\hat{\theta}^* - \hat{\theta})/SE^*$, and inverts its empirical quantiles. It needs a standard error inside every resample, which is expensive, and it is more accurate when the studentized quantity is closer to pivotal than the estimate itself.
- The **BCa interval** adjusts the percentile endpoints for median bias and for an acceleration constant that captures how the standard error changes with the parameter. It is the usual recommendation when the resampling distribution is visibly skewed (Davison & Hinkley, 1997; Efron & Tibshirani, 1993).

For the three targets of Section 6 the skewness values -0.3891 , 0.2635 , and 0.3785 are mild, so percentile and normal-approximation endpoints agree closely and the higher-order corrections would change little. That agreement is a property of this example, not a general licence.

Extended practice: resampling choices (Integrative)

1. Repeat the case bootstrap of Section 6.2 for the women coefficient in `m2`, with the same seed and B . Report the bootstrap SE and percentile interval, compare them with the classical 0.0097 and the HC3 0.0079 , and explain what it means that the classical SE is the largest of the three here. State what the comparison does not show.
2. Resample the STAR observed-reading records in two ways: drawing pupils with replacement within each school, and drawing whole schools with replacement. Report the resulting SEs for the small-minus-regular difference beside the row-level 1.0377 and the clustered 1.8476 . Explain which resampling unit the design supports and why, and say what neither scheme repairs.

References

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>
- Baumann, J. F., Seifert-Kessell, N., & Jones, L. A. (1992). Effect of think-aloud instruction on elementary students' comprehension monitoring abilities. *Journal of Reading Behavior*, 24(2), 143–172. <https://doi.org/10.1080/10862969209547770>
- Chetty, R., Friedman, J. N., Hilger, N., Saez, E., Schanzenbach, D. W., & Yagan, D. (2011). How does your kindergarten classroom affect your earnings? Evidence from Project STAR. *The Quarterly Journal of Economics*, 126(4), 1593–1660.
- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511802843>

- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429246593>
- Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, 27(3), 557–577.
- Fox, J. (2016). *Applied regression analysis and generalized linear models* (3rd ed.). SAGE Publications.
- Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. <https://www.john-fox.ca/Companion/index.html>
- Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.carData>
- Freedman, D. A. (1981). Bootstrapping regression models. *The Annals of Statistics*, 9(6), 1218–1228. <https://doi.org/10.1214/aos/1176345638>
- Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and other stories*. Cambridge University Press. <https://doi.org/10.1017/9781139161879>
- Gelman, A., & Stern, H. (2006). The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4), 328–331. <https://doi.org/10.1198/000313006X152649>
- Hanushek, E. A. (1999). Some findings from an independent investigation of the Tennessee STAR experiment and from other investigations of class size effects. *Educational Evaluation and Policy Analysis*, 21(2), 143–163.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
- Kleiber, C., & Zeileis, A. (2008). *Applied econometrics with R*. Springer. <https://doi.org/10.1007/978-0-387-77318-6>
- Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, 114(2), 497–532.
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.
- Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7)
- Mosteller, F. (1995). The Tennessee study of class size in the early school grades. *The Future of Children*, 5(2), 113–127.
- Nye, B., Hedges, L. V., & Konstantopoulos, S. (2000). The effects of small classes on academic achievement: The results of the Tennessee class size experiment. *American Educational Research Journal*, 37(1), 123–151.

- Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464. <https://doi.org/10.1214/aos/1176344136>
- Stock, J. H., & Watson, M. W. (2007). *Introduction to econometrics* (2nd ed.). Addison Wesley.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838. <https://doi.org/10.2307/1912934>
- Zeileis, A. (2004). Econometric computing with HC and HAC covariance matrix estimators. *Journal of Statistical Software*, 11(10), 1–17. <https://doi.org/10.18637/jss.v011.i10>
- Zeileis, A., Köll, S., & Graham, N. (2020). Various versatile variances: An object-oriented implementation of clustered covariances in R. *Journal of Statistical Software*, 95(1), 1–36. <https://doi.org/10.18637/jss.v095.i01>